Pentagonal Pyramid

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A pentagonal pyramid is pyramid having a pentagonal base. The edge length e and slant height s of a pentagonal pyramid with regular base of side length a are given by

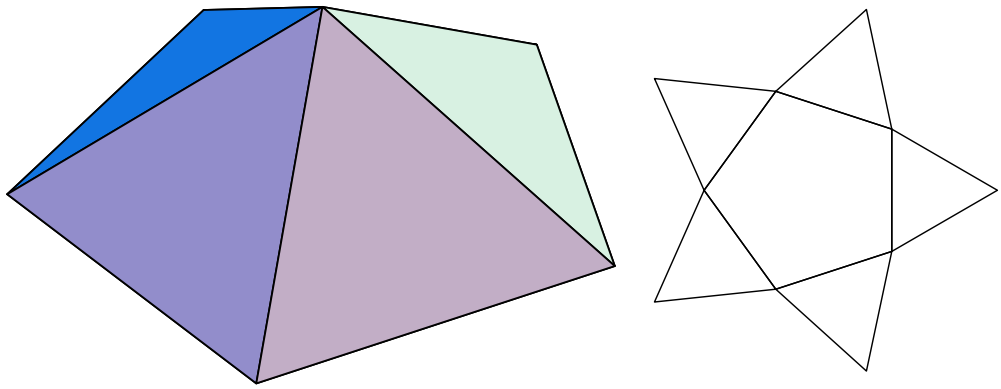
$$e = \sqrt{h^2 + \tfrac{1}{10}\left(5 + \sqrt{5}\right) a^2} \tag{1}$$

$$s = \sqrt{h^2 + \tfrac{1}{20}\left(5 + 2\sqrt{5}\right) a^2}, \tag{2}$$

where h is the height and a is the length of a side of the base. It has surface area and volume

$$S = \frac{5\,a\left(a + \sqrt{a^2 + 4\left(5 - 2\sqrt{5}\right) h^2}\right)}{4\sqrt{5 - 2\sqrt{5}}} \tag{3}$$

$$V = \tfrac{1}{12}\, a^2\, h \sqrt{25 + 10\sqrt{5}}\,. \tag{4}$$



The regular pentagonal pyramid having equilateral triangles as faces so that all its edges are of the same length is Johnson solid J_2.

For the equilateral pentagonal pyramid with edge length a, the slant height is

$$s = \tfrac{1}{2}\sqrt{3}\, a, \tag{5}$$

and the surface area and volume are

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$$V = \frac{1}{24}\left(5+\sqrt{5}\right)a^3. \tag{7}$$

SEE ALSO

Johnson Solid, Pentagon, Pentagonal Prism, Pyramid, Regular Pyramid, Square Pyramid, Triangular Pyramid

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More things to try: = pentagonal pyramid = 0, 1, 3, 7, 15 = circle

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